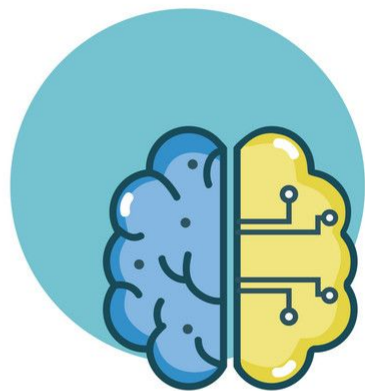


INTRODUCTION TO MACHINE LEARNING

DEEP GENERATIVE MODELS

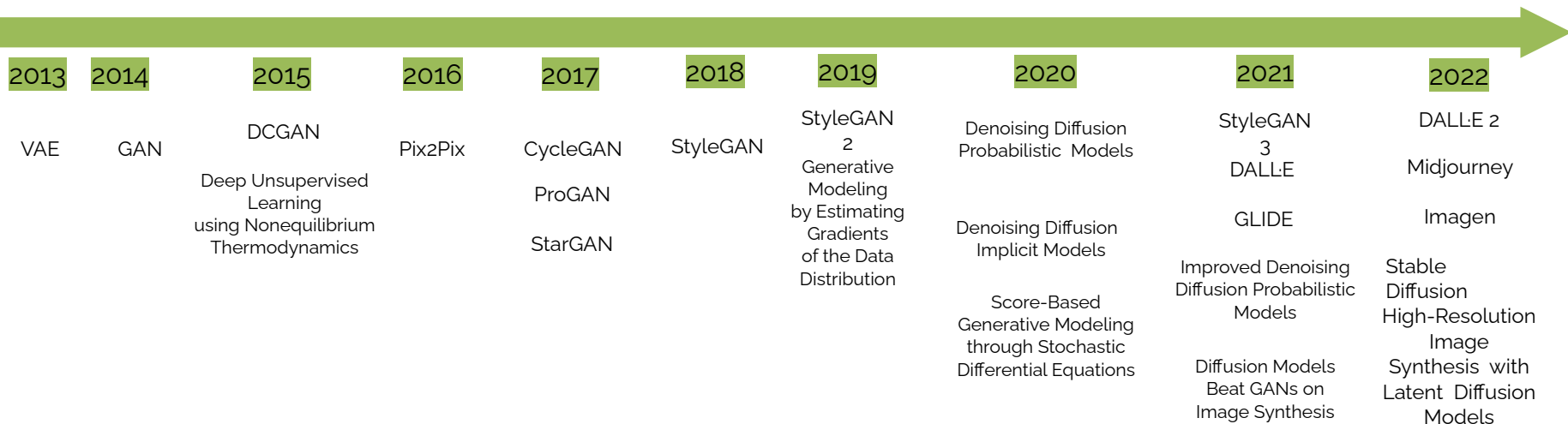


Elisa Ricci





GENERATIVE TIMELINE



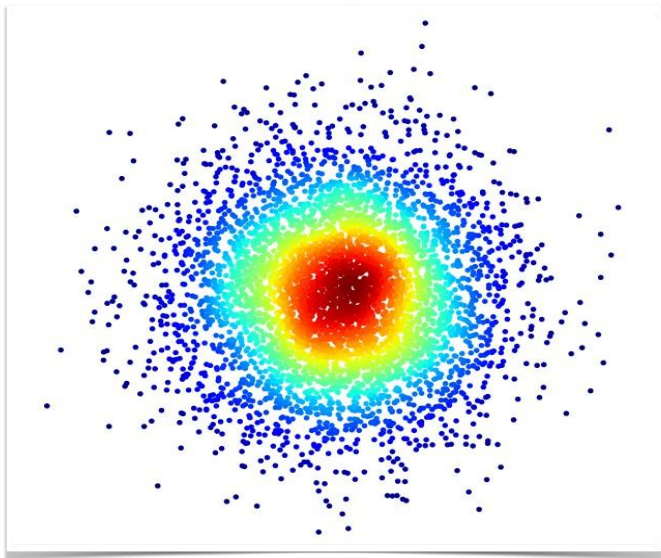
GENERATIVE VS DISCRIMINATIVE MODEL

- **Generative models** are statistical models of the data distribution p_X or p_{XY} depending on the availability of target data
- **Discriminative models** are statistical models of the conditional distribution $p_{Y|X}$ of the target given the input
- A discriminative model can be constructed from a generative model via Bayes rule but not viceversa

$$p_{Y|X}(y | x) = \frac{p_{XY}(x, y)}{\sum_{y'} p_{XY}(x, y')} \quad] \quad p_X(x)$$

DENSITY ESTIMATION (EXPLICIT)

Find a probability distribution $f \in \Delta(\mathcal{Z})$ that fits the data, where $z \in \mathcal{Z}$ is sampled from an unknown data distribution $p_{\text{data}} \in \Delta(\mathcal{Z})$



SUPERVISED
LEARNING

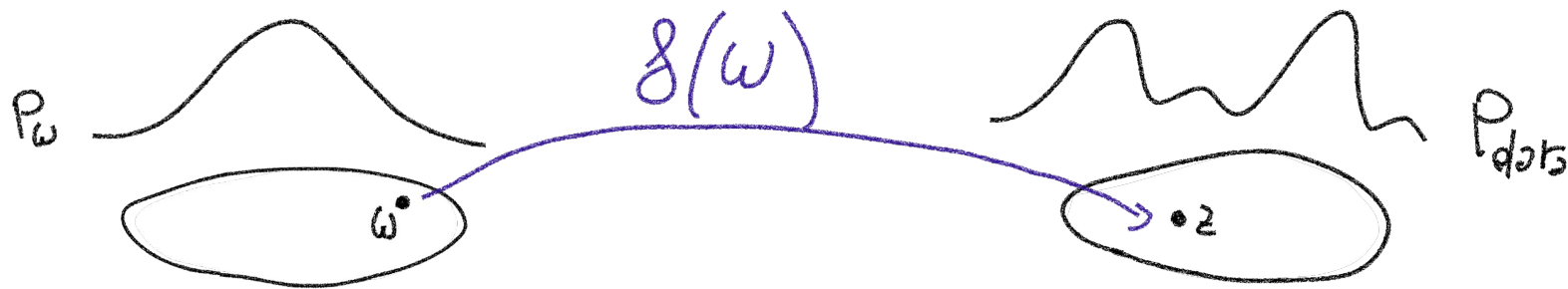
UNSUPERVISED
LEARNING

$$\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$$

$$\mathcal{Z} = \mathcal{X}$$

DENSITY ESTIMATION (IMPLICIT)

Find a function $f \in \mathcal{F}^\Omega$ that generates data $f(\omega) \in \mathcal{Z}$ from an input ω sampled from some predefined distribution $p_\omega \in \Delta(\Omega)$ in a way that the distribution of generated samples fits the (unknown) data distribution $p_{\text{data}} \in \Delta(\mathcal{Z})$



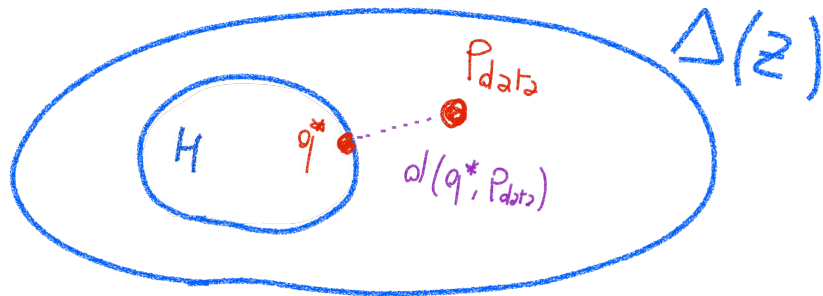
SUPERVISED
LEARNING $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$

UNSUPERVISED
LEARNING $\mathcal{Z} = \mathcal{X}$

OBJECTIVE

- Define an **hypothesis space** $\mathcal{H} \subset \Delta(\mathcal{Z})$ of models that can represent probability distributions (implicitly or explicitly)
- Define a **divergence measure** $d \in \mathbb{R}^{\Delta(\mathcal{Z}) \times \Delta(\mathcal{Z})}$ between probability distributions in $\Delta(\mathcal{Z})$ (e.g. Kullback-Leibler divergence)
- Find an hypothesis $q^* \in \mathcal{H}$ that best fits the data distributed according to p_{data} .
- The best fit is measured using the divergence d , i.e.

$$q^* \in \arg \min_{q \in \mathcal{H}} d(p_{\text{data}}, q)$$

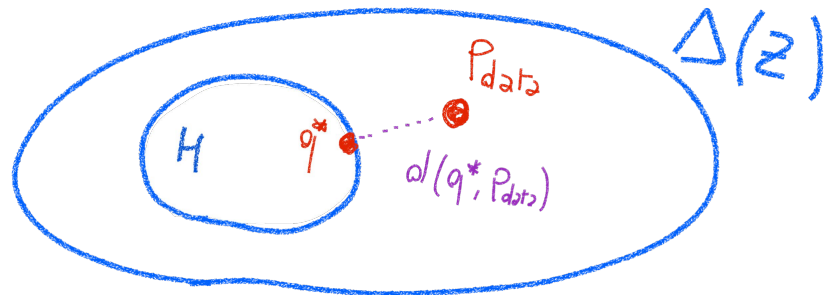


OBJECTIVE

WE ASSUME $\mathcal{Z} = \mathcal{X}$
IN THE REST OF THE LECTURE

- Define an **hypothesis space** $\mathcal{H} \subset \Delta(\mathcal{Z})$ of models that can represent probability distributions (implicitly or explicitly)
- Define a **divergence measure** $d \in \mathbb{R}^{\Delta(\mathcal{Z}) \times \Delta(\mathcal{Z})}$ between probability distributions in $\Delta(\mathcal{Z})$ (e.g. Kullback-Leibler divergence)
- Find an hypothesis $q^* \in \mathcal{H}$ that best fits the data distributed according to p_{data} .
- The best fit is measured using the divergence d , i.e.

$$q^* \in \arg \min_{q \in \mathcal{H}} d(p_{\text{data}}, q)$$



TODAY

EXPLICIT
DENSITY



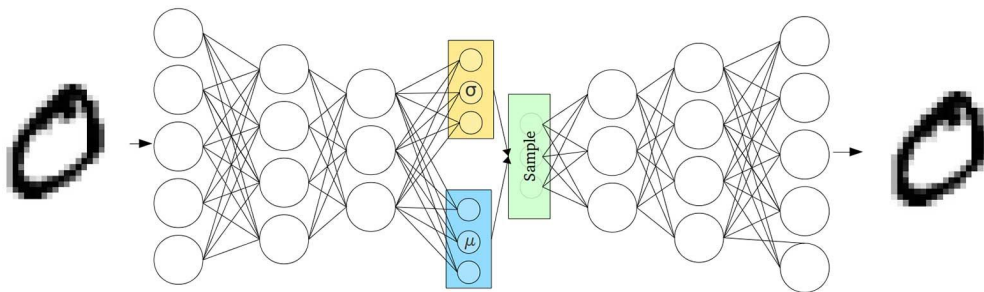
Variational AutoEncoders (VAE)

Generative Adversarial Networks (GAN)

↳ IMPLICIT
DENSITY

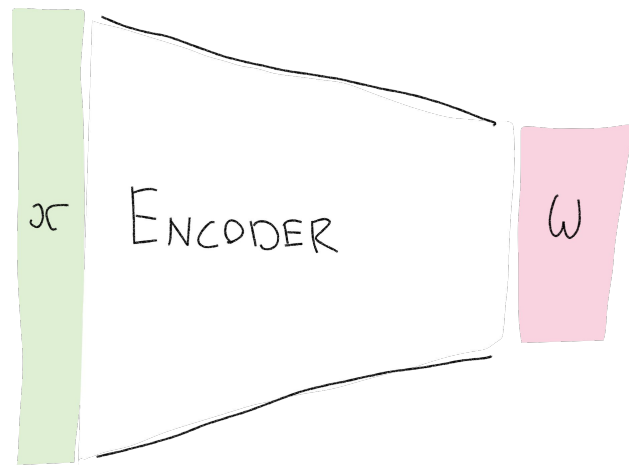
VARIATIONAL AUTOENCODERS (VAE)

- Introduced in 2014 by Kingma and Welling
- Idea: add a probabilistic spin to traditional Autoencoders



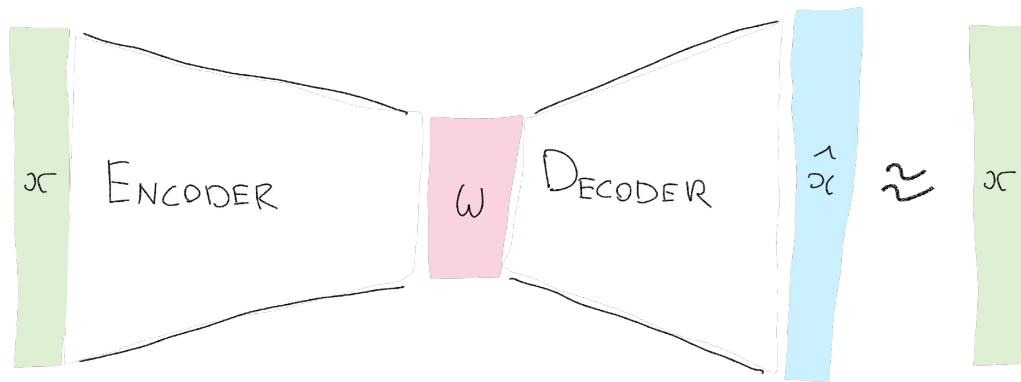
RECAP: AUTOENCODERS

- A way of compressing high-dimensional data into a lower dimensional representation
- An encoder maps input data to a **compressed representation**
- The compressed representation preserves meaningful factors of variations in the data
- E.g. PCA with dimensions dropped can be seen as a linear encoder preserving as much variance in the data as possible



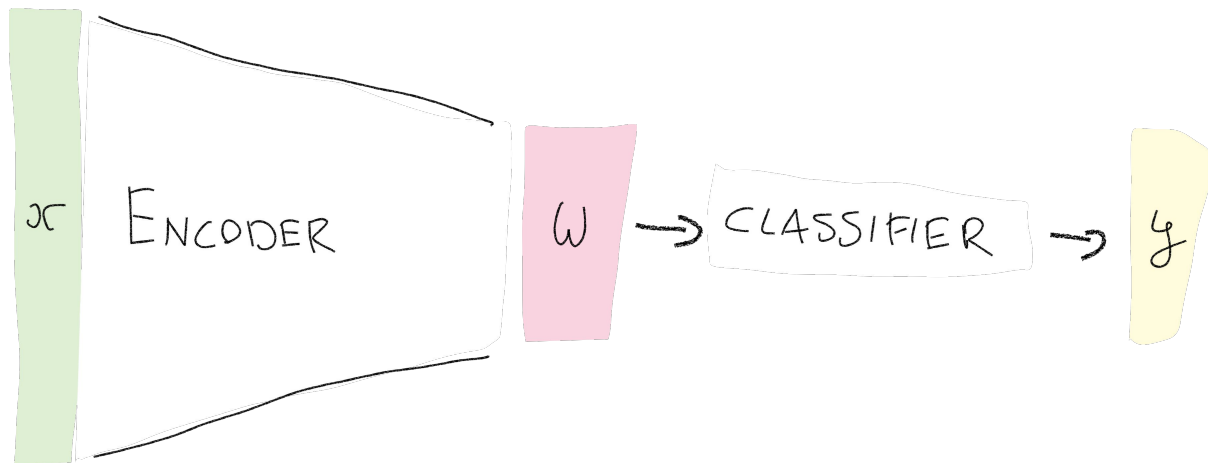
RECAP: AUTOENCODERS

- An encoder is trained by leveraging a decoder mapping the representation back to the input domain yielding $\hat{x}$ (reconstruction)
- So the encoder “autoencodes” its input
- The objective is to minimize the divergence between input and its reconstruction



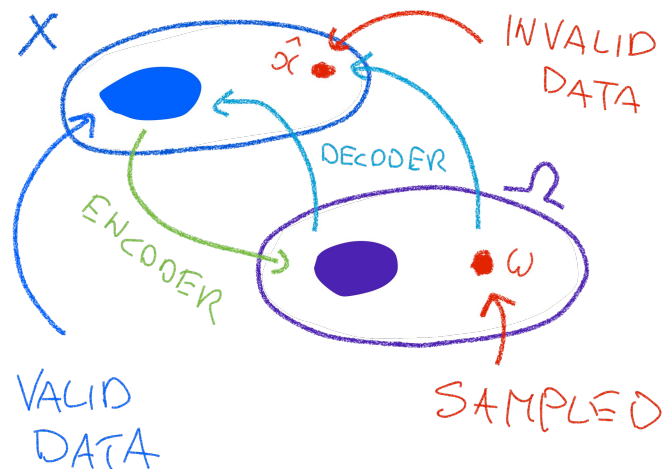
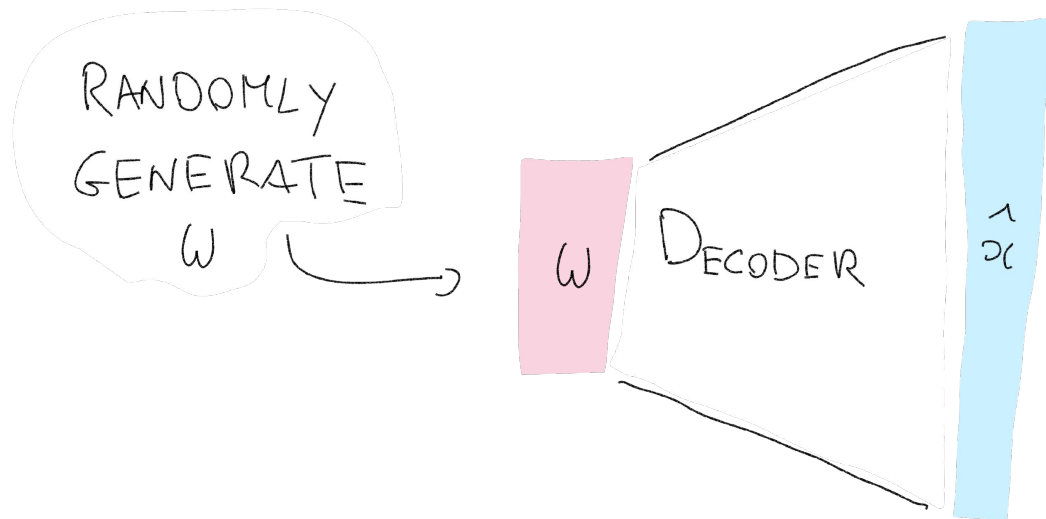
RECAP: AUTOENCODERS

- After training the decoder is not needed anymore, since it was only functional to estimating the encoder
- The encoder can be used to initialize or precompute features for supervised models

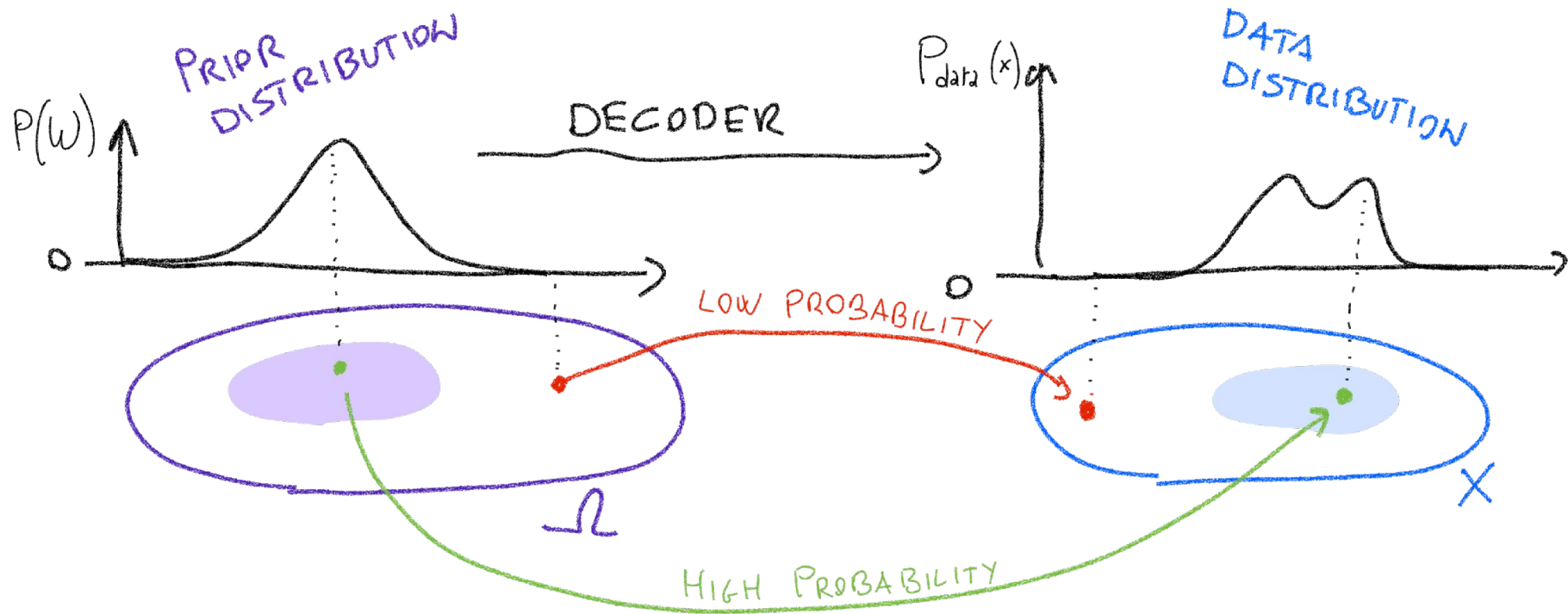


IS THE AUTOENCODER A GENERATIVE MODEL?

- The decoder could be used to generate new data !
- But it will **not** generate data according to the data distribution p_{data}



WHAT WE WOULD LIKE



IN FORMULAS

$$q_{\theta}(x) = \mathbb{E}_{\omega \sim p^{\omega}} [q_{\theta}(x | \omega)]$$

Diagram annotations:

- PARAMETERS** (green text) with an arrow pointing to θ in $q_{\theta}(x)$.
- PRIOR** (blue text) with an arrow pointing to $\omega \sim p^{\omega}$.
- DECODER** (red text) with an arrow pointing to $q_{\theta}(x | \omega)$.

OBJECTIVE

$$\theta^{\star} \in \arg \min_{\theta \in \Theta} d(q_{\theta}, p_{\text{data}})$$

WE WILL USE THE
KL-DIVERGENCE

$$d_{\text{KL}}(p, q) = \mathbb{E}_{x \sim p} \left[\log \frac{p(x)}{q(x)} \right]$$

INTRACTABLE OBJECTIVE

$$\begin{aligned}d_{\text{KL}}(p_{\text{data}}, q_{\theta}) &= \mathbb{E}_{x \sim p_{\text{data}}} \left[\log \frac{p_{\text{data}}(x)}{q_{\theta}(x)} \right] \\&= - \mathbb{E}_{x \sim p_{\text{data}}} [\log q_{\theta}(x)] + \text{const} \\&= - \mathbb{E}_{x \sim p_{\text{data}}} \left[\log \mathbb{E}_{\omega \sim p_{\omega}} [q_{\theta}(x | \omega)] \right] + \text{const}\end{aligned}$$

INTRACTABLE



INTRACTABLE OBJECTIVE

Let $q_\psi(\omega | x) \in \Delta(\Omega)$ denote an encoding probability distribution

↪ OTHER PARAMETERS

$$\log \mathbb{E}_{\omega \sim p_\omega} [q_\theta(x | \omega)] = \log \mathbb{E}_{\omega \sim q_\psi(\cdot | x)} \left[q_\theta(x | \omega) \frac{p_\omega(\omega)}{q_\psi(\omega | x)} \right]$$

by JENSEN'S
INEQUALITY

$$\geq \mathbb{E}_{\omega \sim q_\psi(\cdot | x)} \left[\log \left(q_\theta(x | \omega) \frac{p_\omega(\omega)}{q_\psi(\omega | x)} \right) \right]$$

$$= \mathbb{E}_{\omega \sim q_\psi(\cdot | x)} [\log q_\theta(x | \omega)] - d_{\text{KL}}(q_\psi(\cdot | x), p_\omega)$$

RECONSTRUCTION

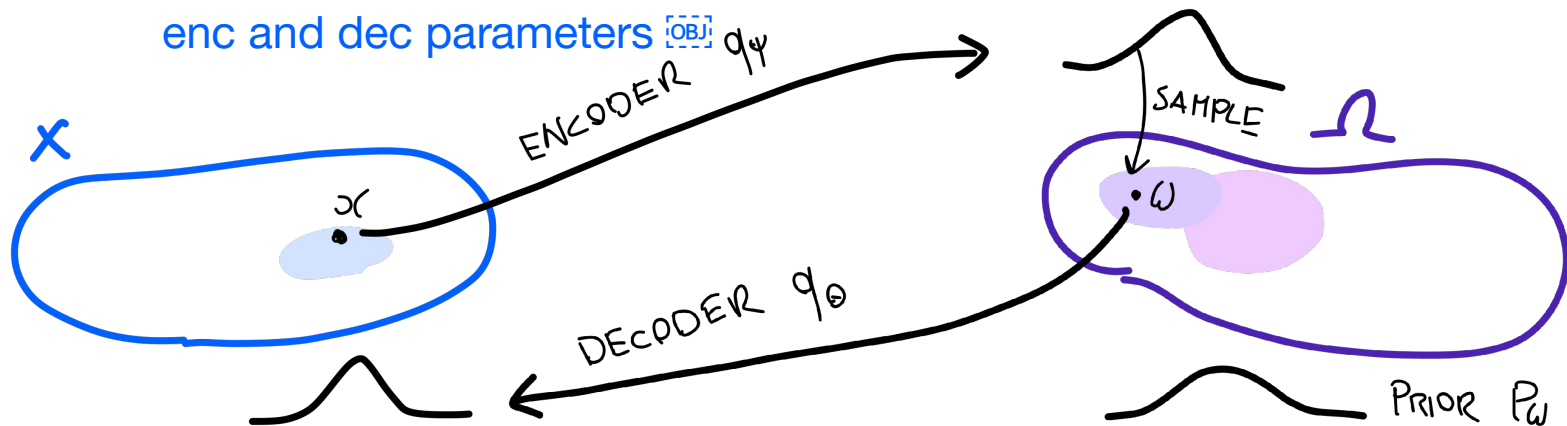
REGULARIZER

VARIATIONAL BOUND

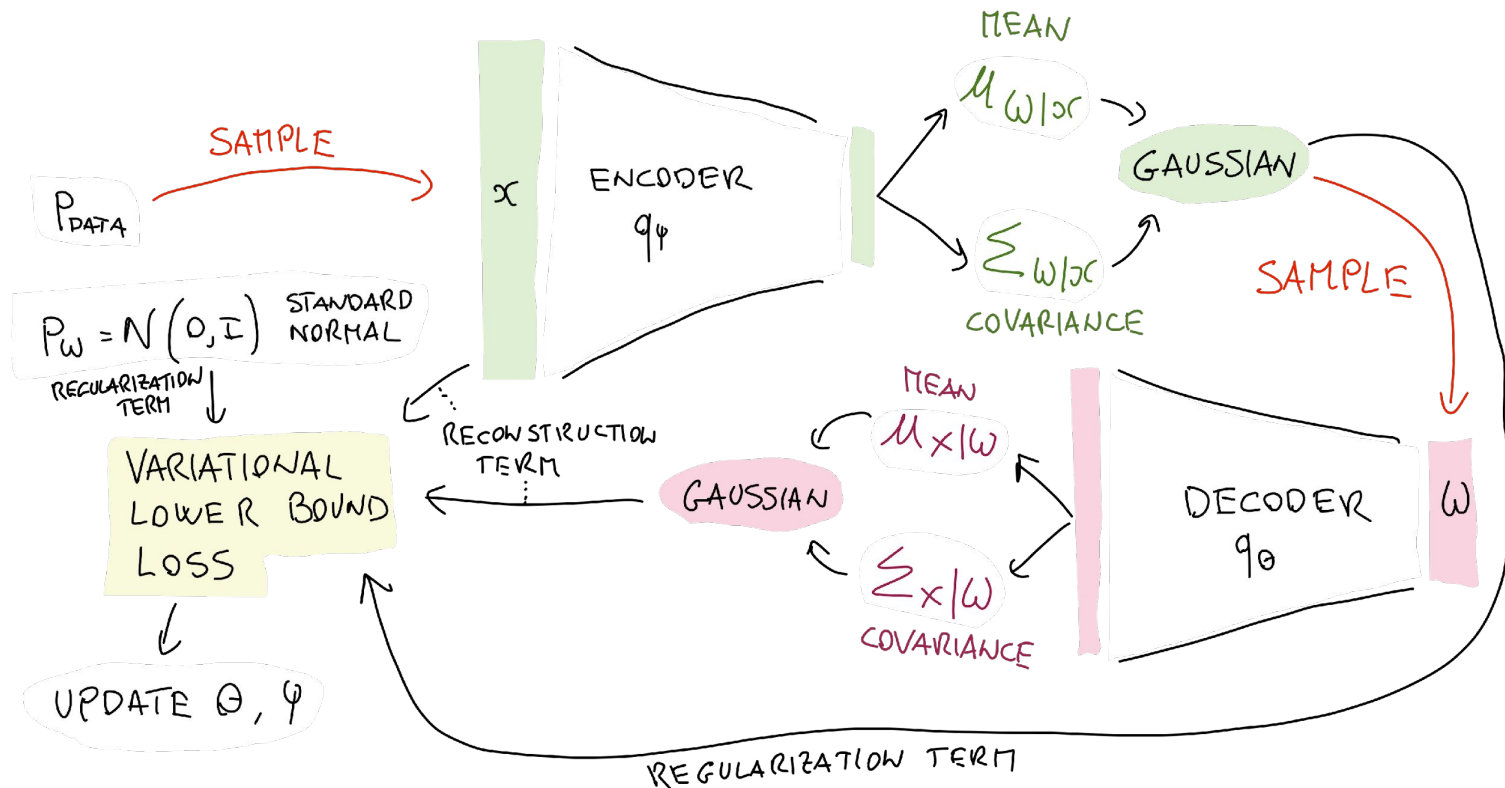
$$d_{\text{KL}}(p_{\text{data}}, q_{\theta}) \leq \mathbb{E}_{x \sim p_{\text{data}}} \left[\underbrace{-\mathbb{E}_{\omega \sim q_{\psi}(\cdot | x)} [\log q_{\theta}(x | \omega)]}_{\text{blue wavy line}} + \underbrace{d_{\text{KL}}(q_{\psi}(\cdot | x), p_{\omega})}_{\text{red wavy line}} \right] + \text{const}$$

- Still intractable to compute but we can estimate the gradients of enc and dec parameters ^[OBJ] q_{ψ}

- It might have closed-form solution (e.g. using Gaussian distributions)



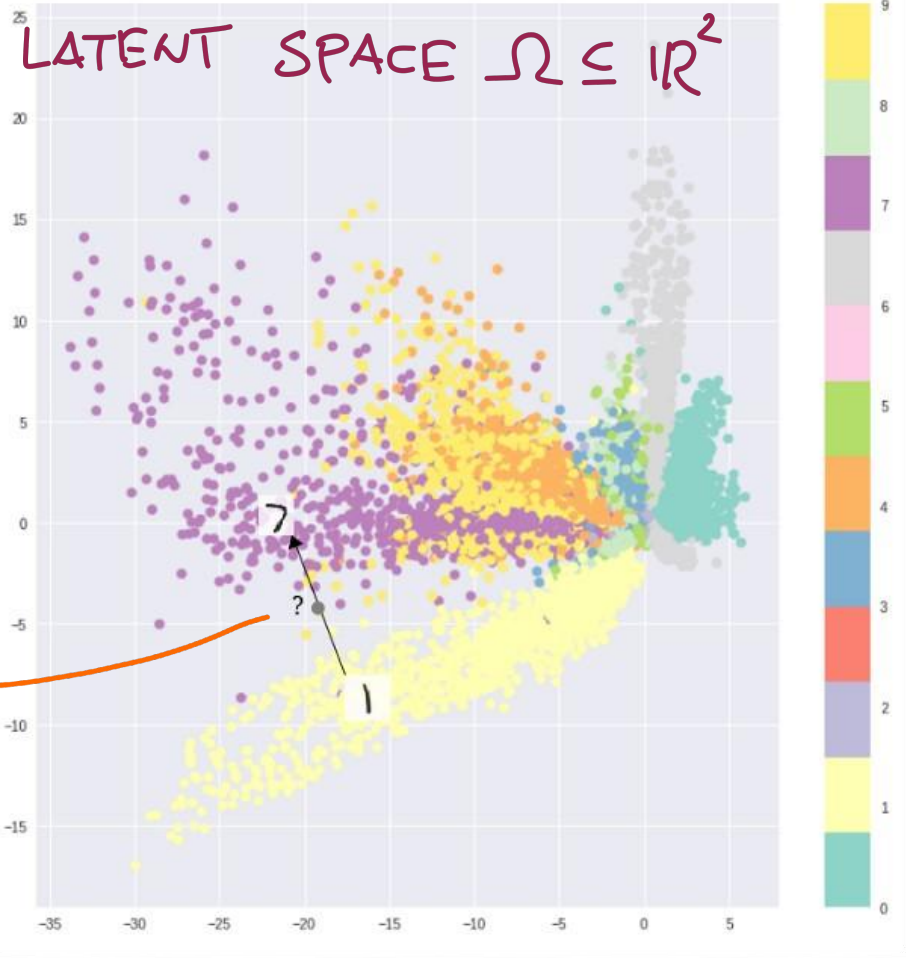
TRAINING IN PRACTICE



EXAMPLES OF LEARNED LATENT SPACE ON MNIST

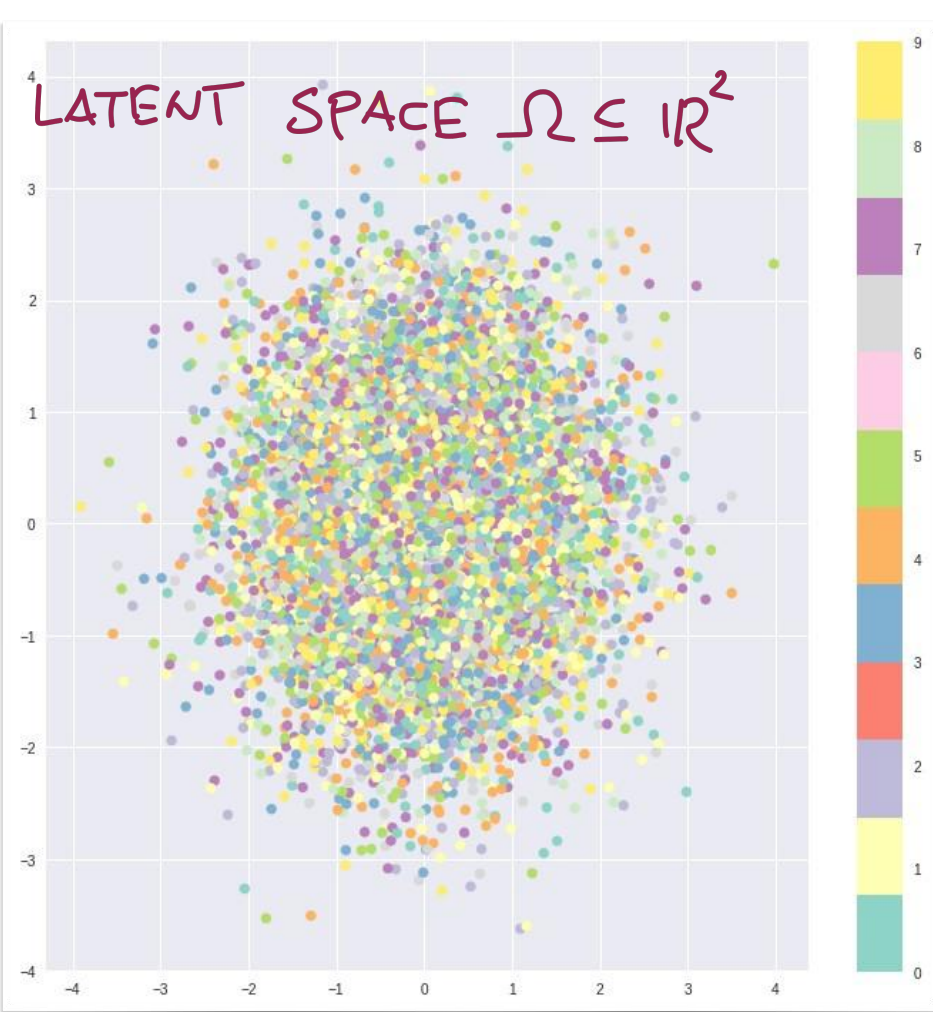
USING ONLY
RECONSTRUCTION LOSS

THIS MIGHT
GENERATE
MEANINGLESS
DATA !

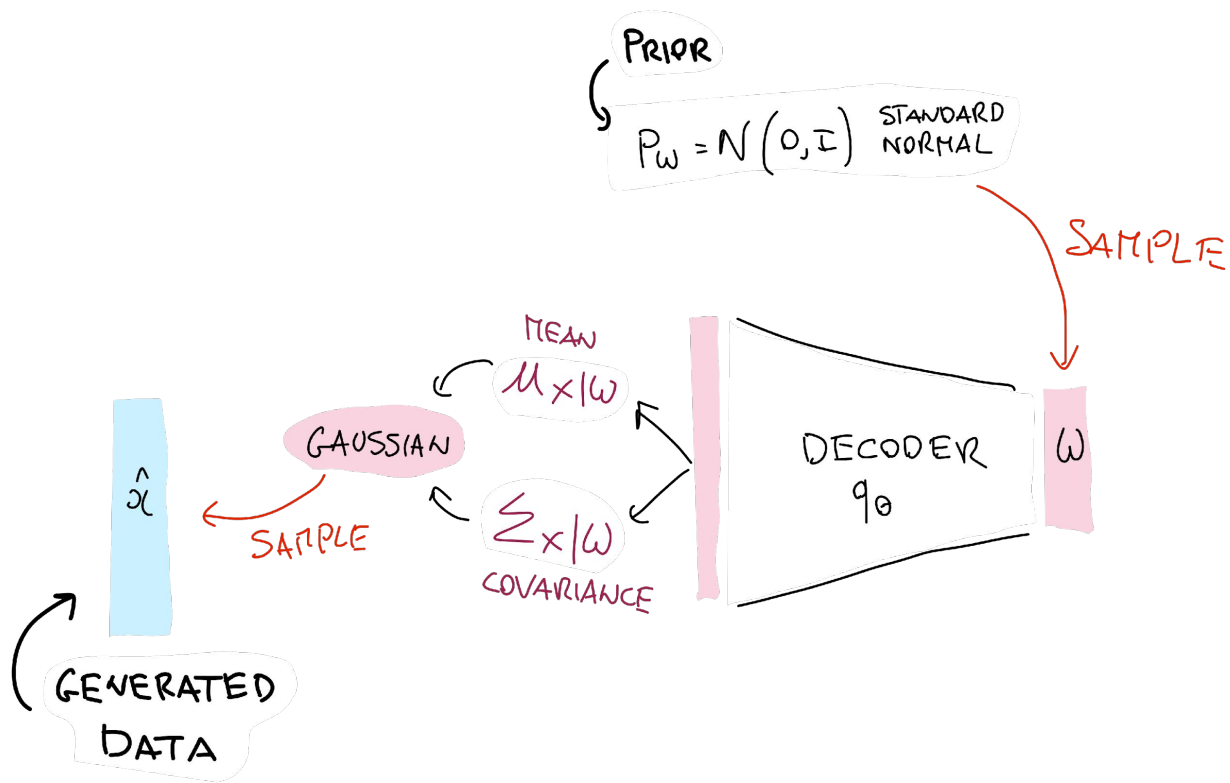


EXAMPLES OF LEARNED LATENT SPACE ON MNIST

USING ONLY
REGULARIZATION
LOSS



GENERATING DATA



EXAMPLES

EXAMPLES



LABELLED FACES IN THE WILD

32x32 CIFAR-10

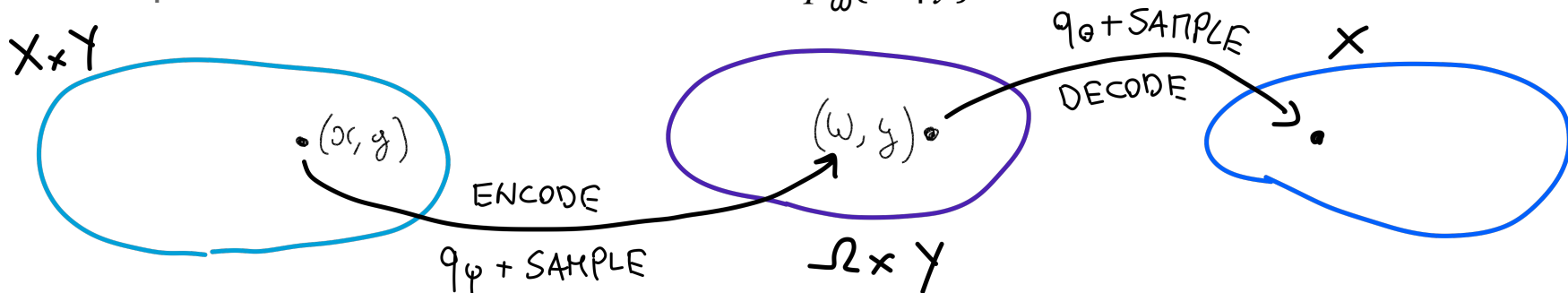


CONDITIONAL VAE

Assume we have side information $y \in \mathcal{Y}$ (e.g. digit labels, attributes, etc.) and we want to generate new data conditioned on the side information (e.g. generate digit 7, or generate a face with glasses)

Modify the encoder and decoder to take the side information in input obtaining $q_{\psi}(\omega | x, y)$ and $q_{\theta}(x | \omega, y)$

Define priors conditioned on side information $p_{\omega}(\omega | y)$



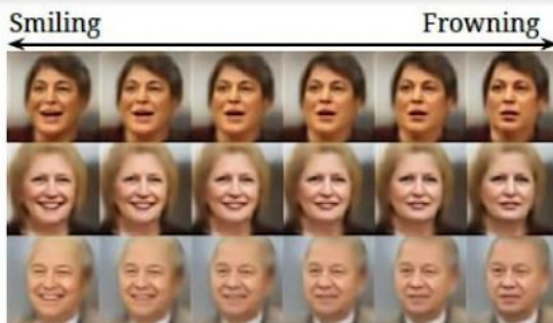
EXAMPLE



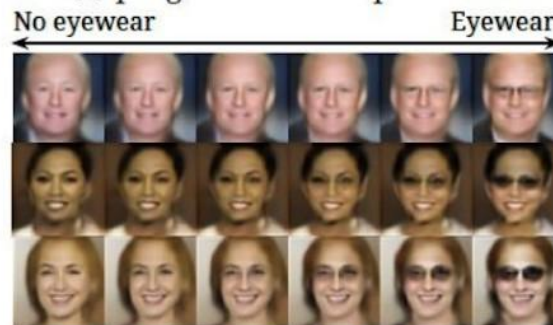
(a) progression on gender



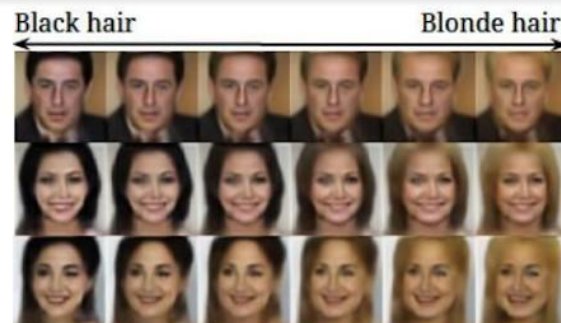
(b) progression on age



(c) progression on expression



(d) progression on eyewear



(e) progression on hair color



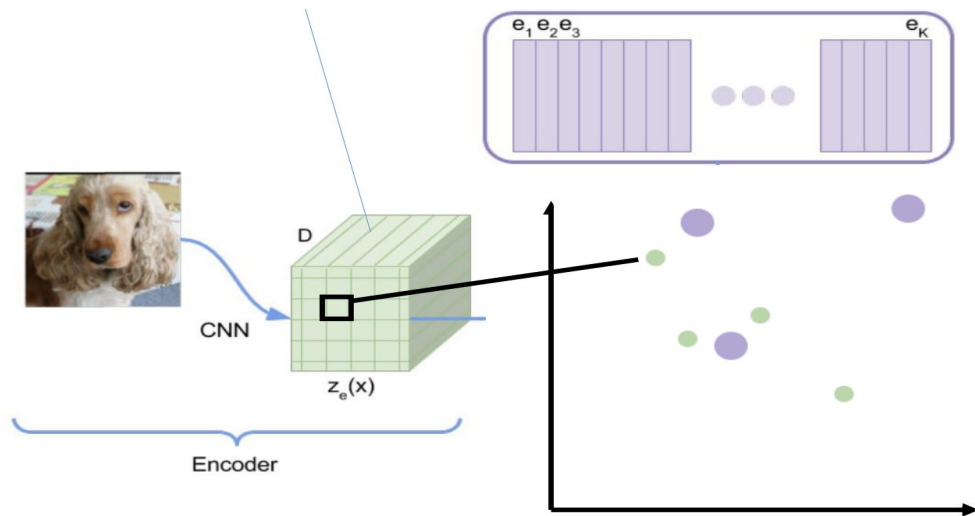
(f) progression on primary color

ISSUES WITH VAE

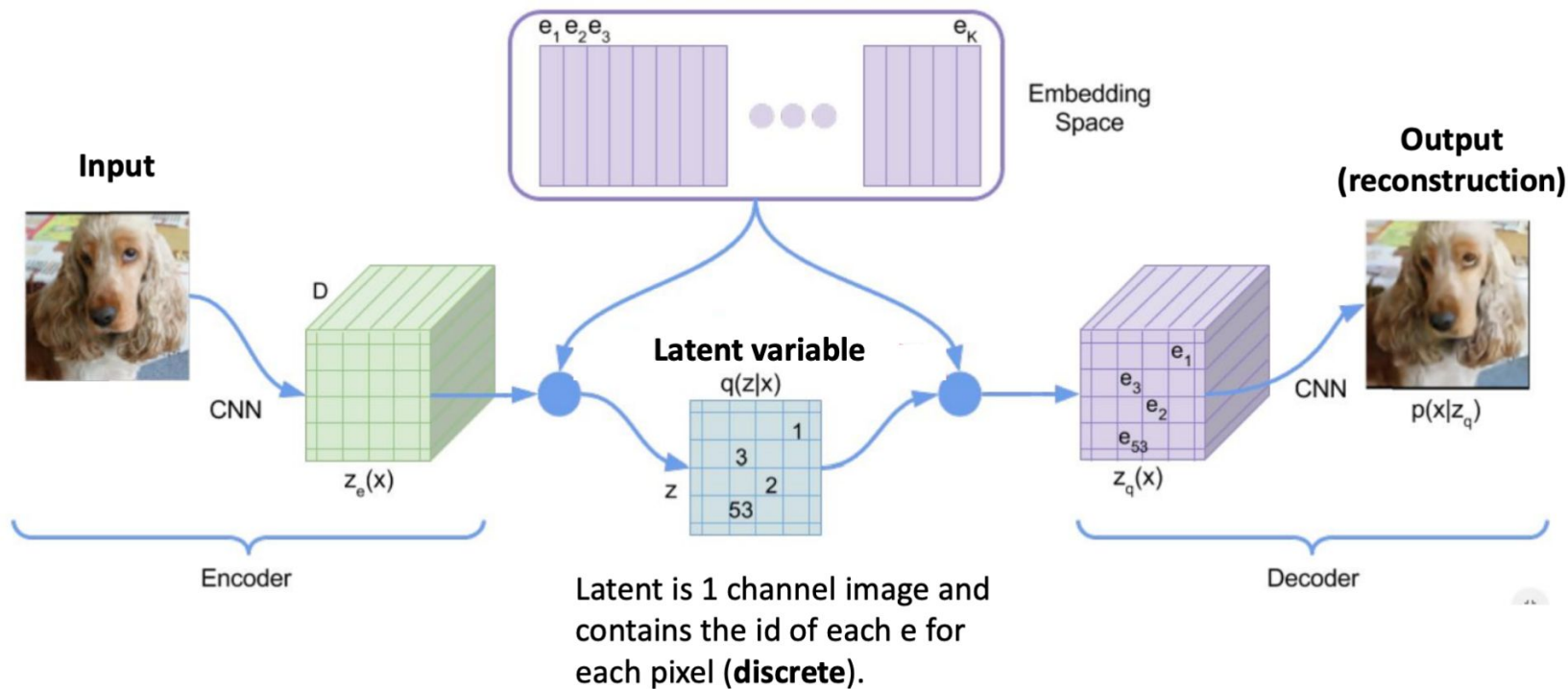
- **Underfitting:** at initial stages the regularizer is too strong and tends to annihilate the model capacity
- **Blurry samples:** the generator tends to produce blurry data

NEW MODELS WITH VAE: VQ-VAE

- VQ-VAE-2 is a image synthesis model based on Variational Autoencoders. It produces images that are high quality comparable to big GANs models
- Discrete space with Dictionary



VQ-VAE



VQ-VAE

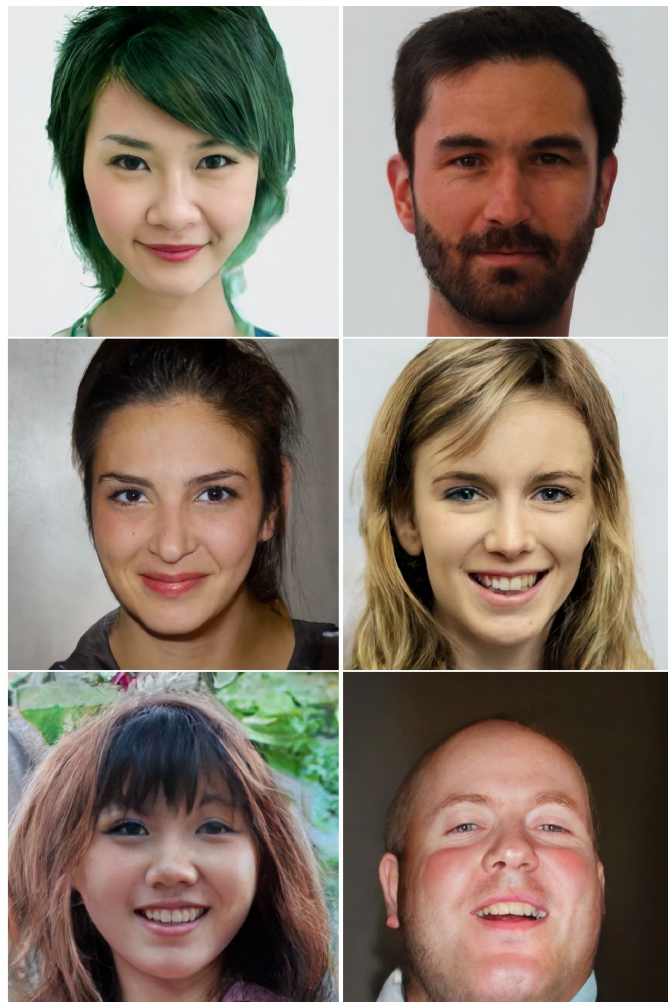
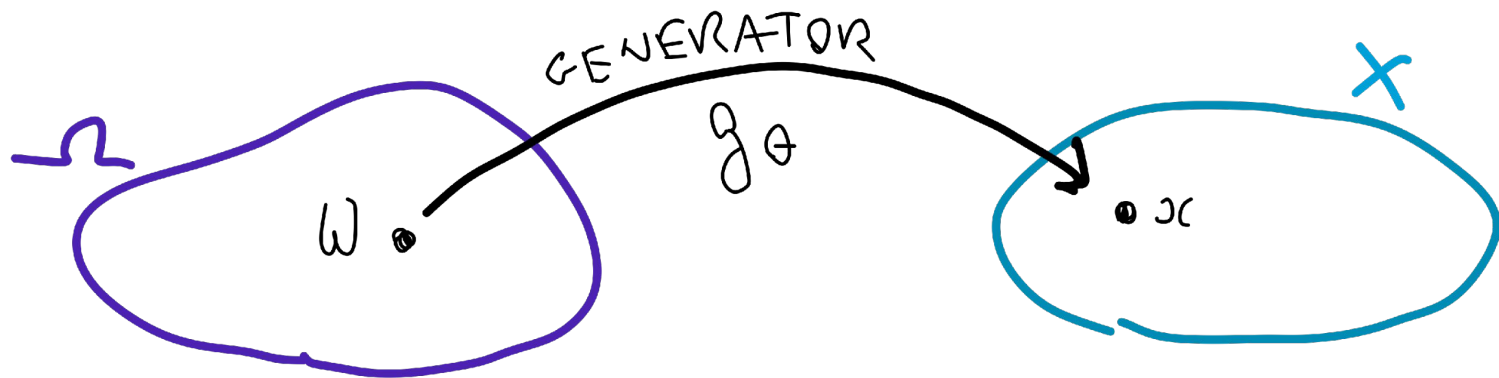


Figure 6: Representative samples from the three layer hierarchical model trained on FFHQ 1094 v

GENERATIVE ADVERSARIAL NETWORKS (GAN)

GANs enable the possibility of estimating implicit densities

We assume to have a prior density $p_\omega \in \Delta(\Omega)$ given and a generator (or decoder) $g_\theta \in \mathcal{G}^\Omega$ that generates data points in $\mathcal{X}$ given a random element from Ω



GENERATIVE ADVERSARIAL NETWORKS (GAN)

- The density induced by the prior p_ω and the generator g_θ is given by $q_\theta(x) = \mathbb{E}_{\omega \sim p_\omega} \delta[g_\theta(\omega) - x]$, where δ is the Dirac delta function.
- The (original) GAN objective is to find $\theta^\star$ such that $q_{\theta^\star}$ best fits the data distribution p_{data} under the Jensen-Shannon divergence:

$$\theta^\star \in \arg \min_{\theta} d_{\text{JS}}(p_{\text{data}}, q_\theta)$$

- INTRACTABLE TO COMPUTE
- SAME HOLDS FOR THE GRADIENTS

where $d_{\text{JS}}(p, q) = \frac{1}{2} d_{\text{KL}}\left(p, \frac{p+q}{2}\right) + \frac{1}{2} d_{\text{KL}}\left(q, \frac{p+q}{2}\right)$

AN EQUIVALENT FORM OF THE JS DIVERGENCE

$$\begin{aligned}d_{\text{JS}}(p, q) &= \frac{1}{2}d_{\text{KL}}\left(p, \frac{p+q}{2}\right) + \frac{1}{2}d_{\text{KL}}\left(q, \frac{p+q}{2}\right) \\&= \frac{1}{2}\mathbb{E}_{x \sim p}\left[\log \frac{2p(x)}{p(x)+q(x)}\right] + \frac{1}{2}\mathbb{E}_{x \sim q}\left[\log \frac{2q(x)}{p(x)+q(x)}\right] \\&= \frac{1}{2}\mathbb{E}_{x \sim p}\left[\log \frac{p(x)}{p(x)+q(x)}\right] + \frac{1}{2}\mathbb{E}_{x \sim q}\left[\log \frac{q(x)}{p(x)+q(x)}\right] + \log(2) \\&= \log(2) + \frac{1}{2}\max_t \left\{ \mathbb{E}_{x \sim p} [\log t(x)] + \mathbb{E}_{x \sim q} [\log(1-t(x))] \right\}\end{aligned}$$

$t(x)$ IS LIKE A BINARY CLASSIFIER

PREDICTING WHETHER x CAME FROM p OR q

IDEA!

LEARN $t(x)$

GAN OBJECTIVE

Let $t_\varphi(x)$ be a classifier (or discriminator) for data point in the training set, we get the following lower bound on our objective

$$\begin{aligned} d_{\text{JS}}(p_{\text{data}}, q_\theta) &= \log(2) + \frac{1}{2} \max_t \left\{ \mathbb{E}_{x \sim p_{\text{data}}} [\log t(x)] + \mathbb{E}_{x \sim q_\theta} [\log(1 - t(x))] \right\} \\ &\geq \log(2) + \frac{1}{2} \max_\varphi \left\{ \mathbb{E}_{x \sim p_{\text{data}}} [\log t_\varphi(x)] + \mathbb{E}_{x \sim q_\theta} [\log(1 - t_\varphi(x))] \right\} \end{aligned}$$

CAN BE NEGLECTED
NOT CHANGING
MINIMIZER θ^*

Which is minimized to obtain the generator's parameters:

$$\theta^* \in \operatorname{argmin}_\theta \max_\varphi \left\{ \mathbb{E}_{x \sim p_{\text{data}}} [\log t_\varphi(x)] + \mathbb{E}_{x \sim q_\theta} [\log(1 - t_\varphi(x))] \right\}$$

GAN OBJECTIVE

$$\theta^{\star} \in \operatorname{argmin}_{\theta} \max_{\varphi} \left\{ \mathbb{E}_{x \sim p_{\text{data}}} \left[\log t_{\varphi}(x) \right] + \mathbb{E}_{x \sim q_{\theta}} \left[\log(1 - t_{\varphi}(x)) \right] \right\}$$

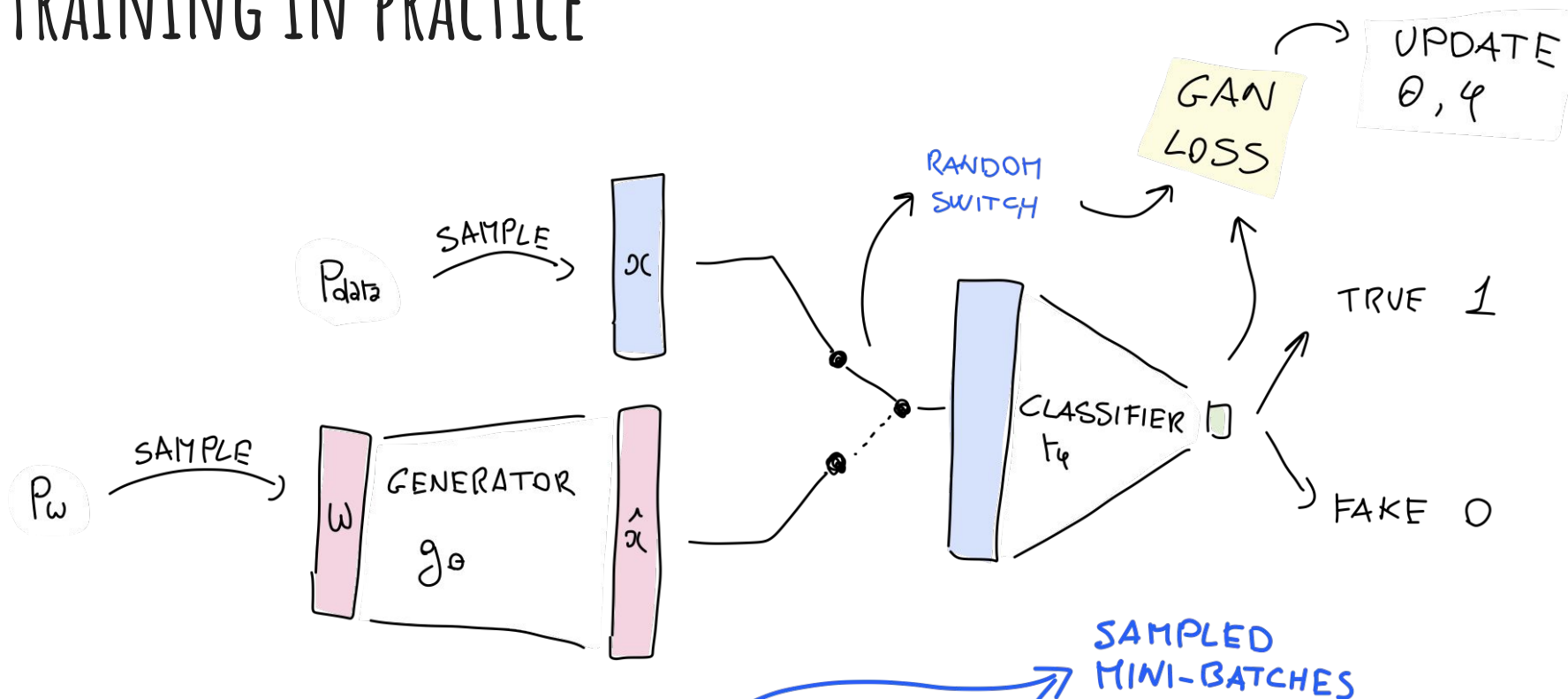
We have the issue that this still depends on the explicit density, however we can equivalently rewrite it as (proof skipped):

$$\theta^{\star} \in \operatorname{argmin}_{\theta} \max_{\varphi} \left\{ \mathbb{E}_{x \sim p_{\text{data}}} \left[\log t_{\varphi}(x) \right] + \mathbb{E}_{\omega \sim p_{\omega}} \left[\log(1 - t_{\varphi}(g_{\theta}(\omega))) \right] \right\}$$

GAN OBJECTIVE

- Two-players, zero-sum, non-cooperative game
- **Player 1:** Generator tries to generate data that cannot be distinguished from the true data
- **Player 2:** Discriminator tries to guess whether the input comes from the true distribution or it is “fake”
- Solved with gradient descent and specialized approaches for handling the min-max problem

TRAINING IN PRACTICE

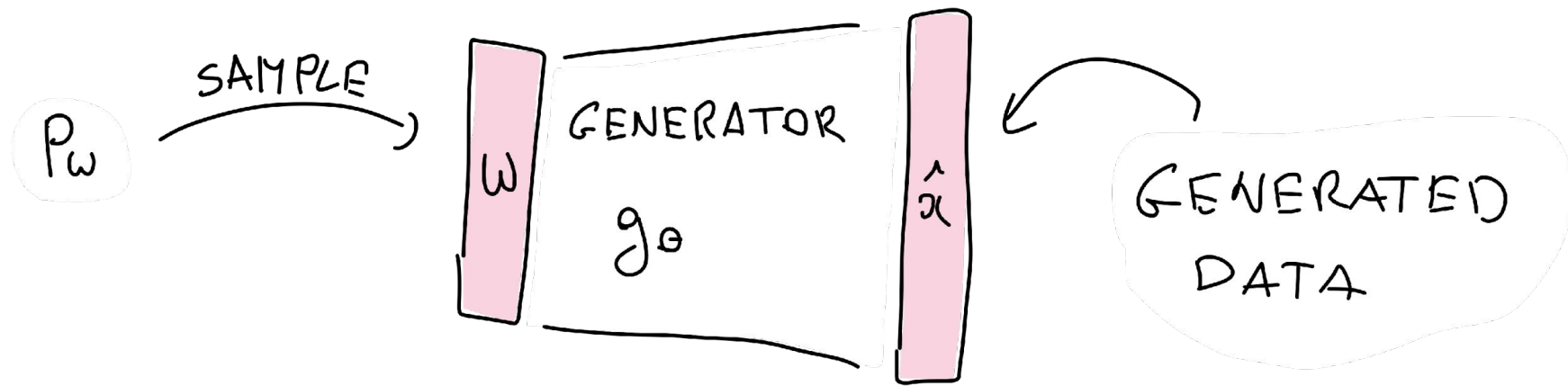


$$\theta^* \in \operatorname{argmin}_{\theta} \max_{\phi} \left\{ \mathbb{E}_{x \sim p_{\text{data}}} \left[\log t_{\phi}(x) \right] + \mathbb{E}_{w \sim p_w} \left[\log(1 - t_{\phi}(g_{\theta}(w))) \right] \right\}$$

Diagrammatic annotations for the equation above:

- A blue arrow points from the term $\mathbb{E}_{x \sim p_{\text{data}}}$ to the text **SAMPLED MINI-BATCHES**.
- A blue arrow points from the term $\mathbb{E}_{w \sim p_w}$ to the text **SAMPLED MINI-BATCHES**.

TEST TIME



EXAMPLE

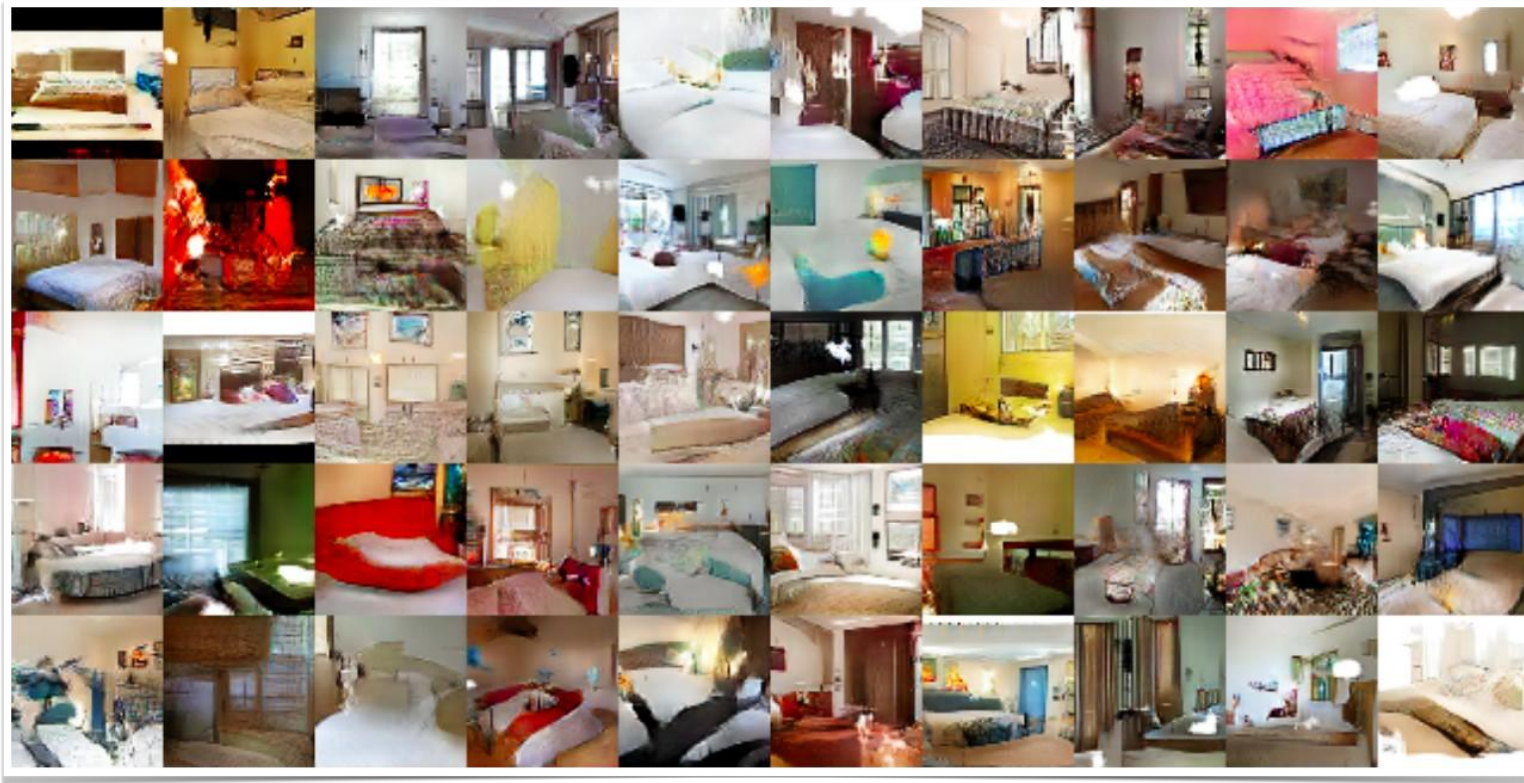
AFTER 1
EPOCH



A. Radford, L. Metz, S. Chintala, Unsupervised representation learning with deep convolutional generative adversarial networks, ICLR 2016

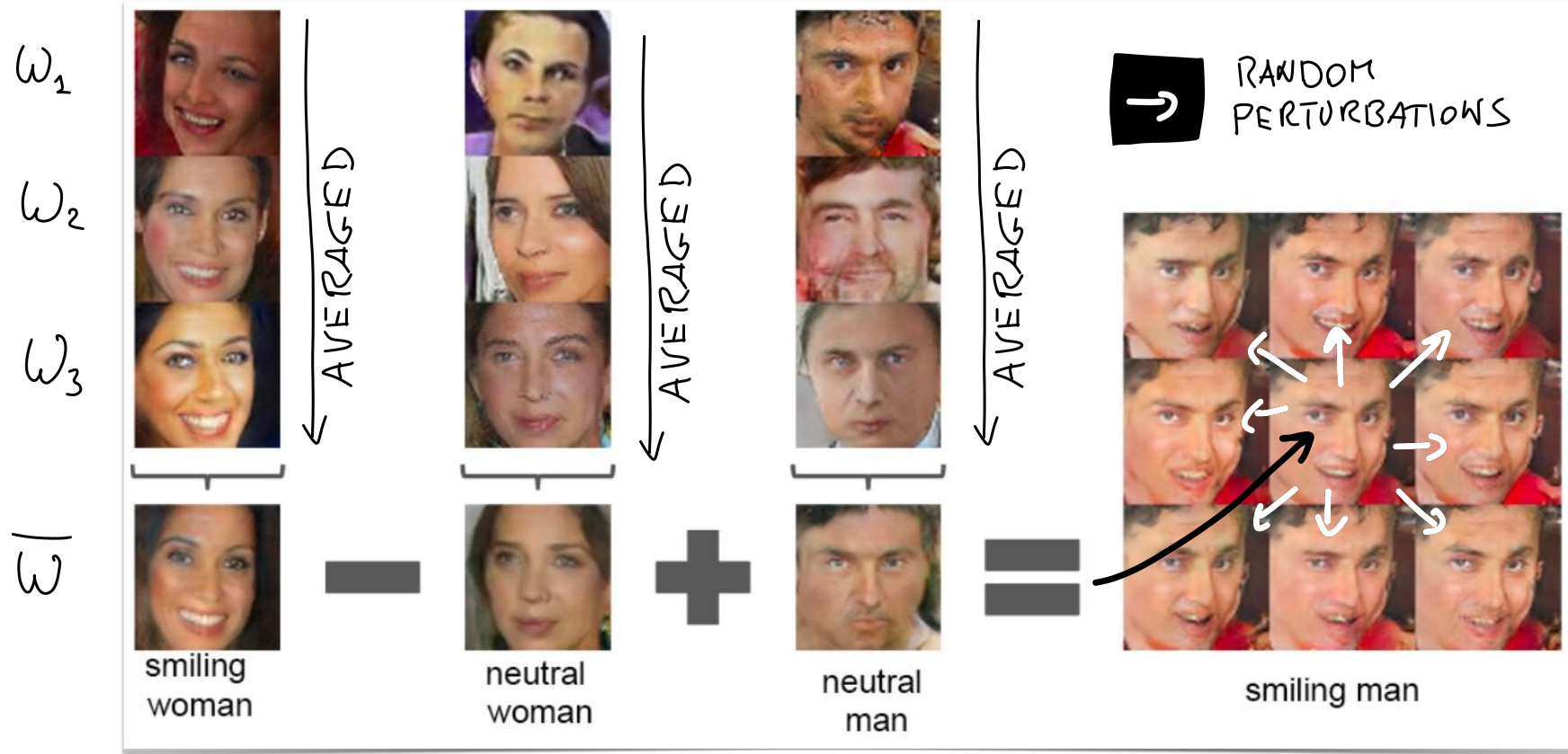
EXAMPLE

AFTER 5
EPOCHS

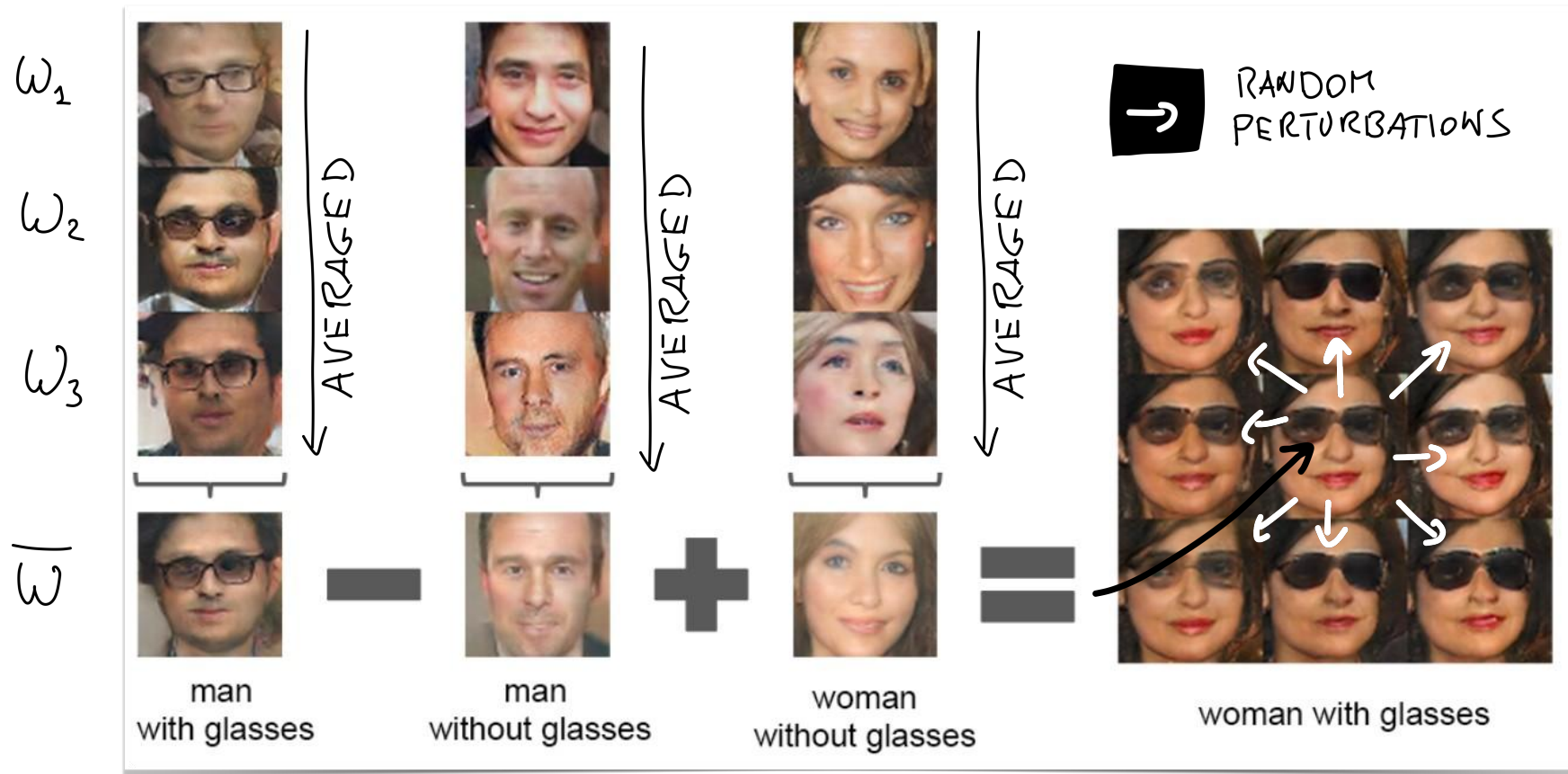


A. Radford, L. Metz, S. Chintala, Unsupervised representation learning with deep convolutional generative adversarial networks, ICLR 2016

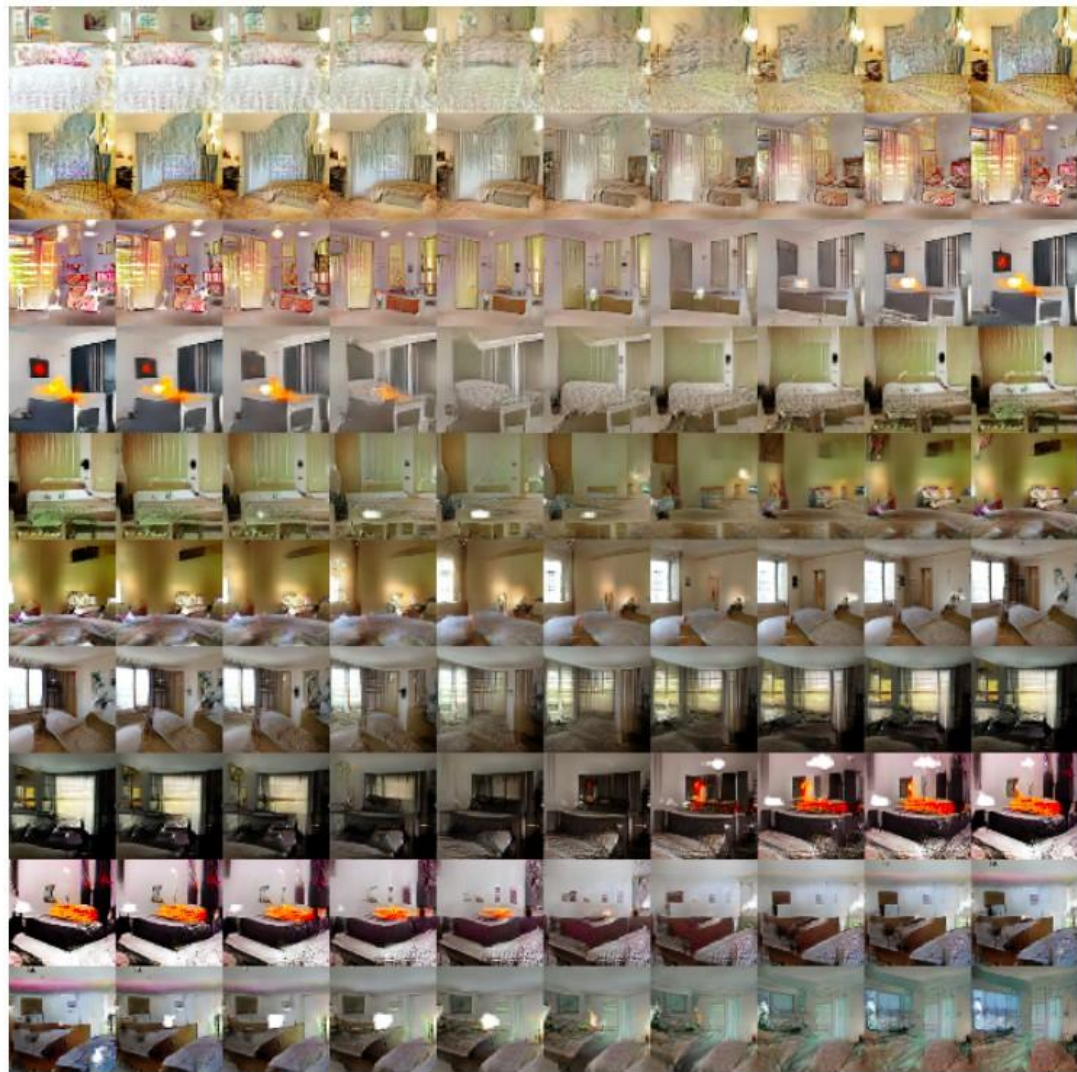
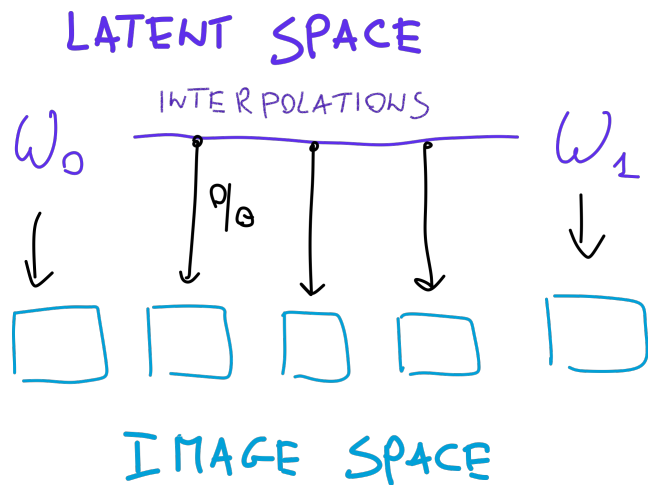
INTERPOLATION IN THE LATENT SPACE



INTERPOLATION IN THE LATENT SPACE

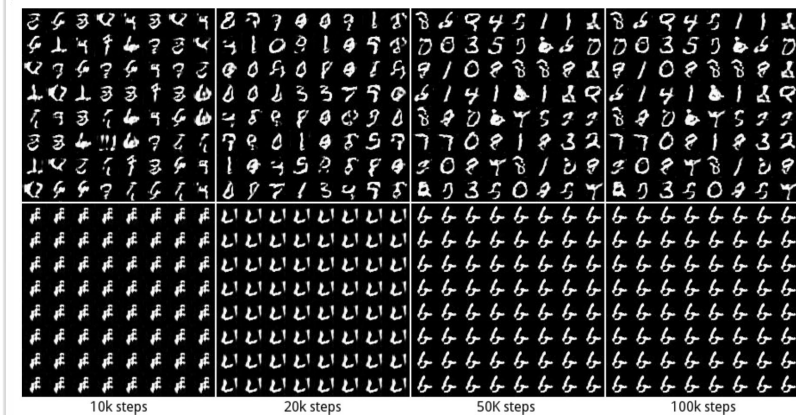
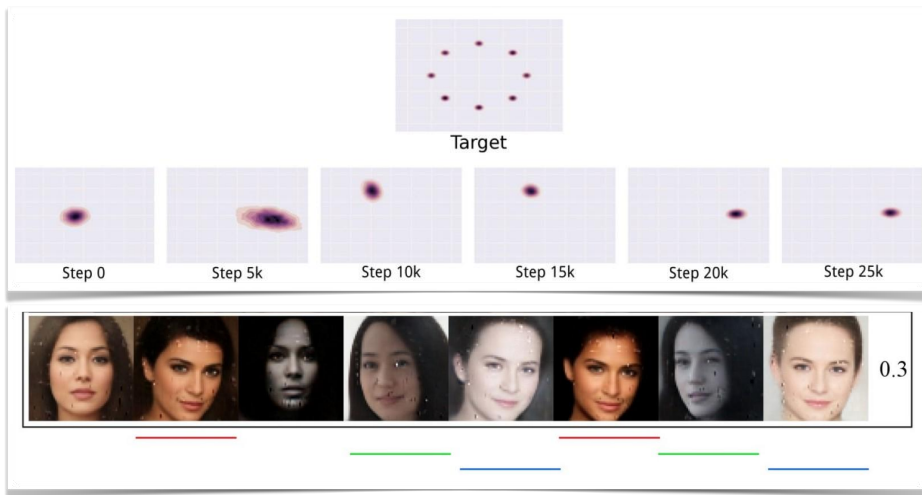


INTERPOLATION IN THE LATENT SPACE



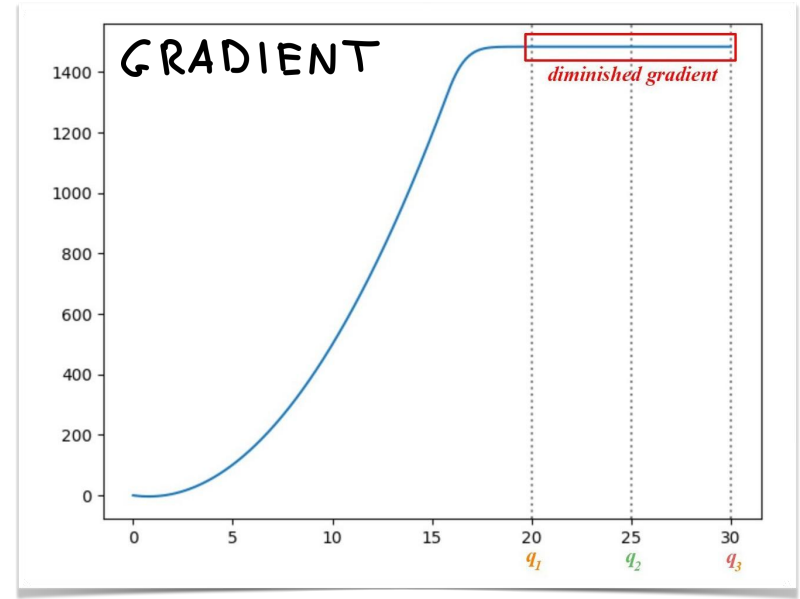
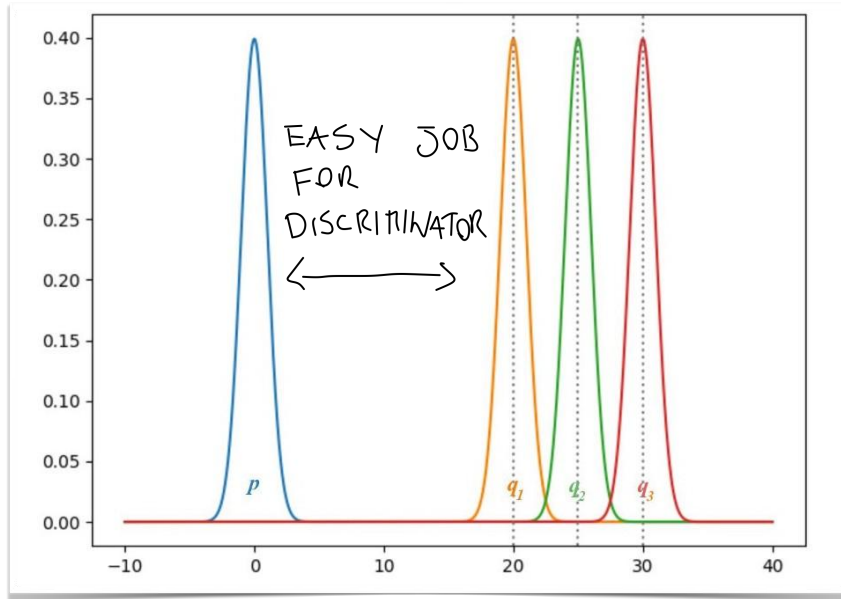
ISSUES WITH GANS

- **Training stability:** parameters might oscillate and never converge
- **Mode collapse:** the generator might learn to perfectly generate few examples from the training set



ISSUES WITH GANS

Vanishing gradient: if the discriminator is very successful it leaves the generator with little gradient to learn from

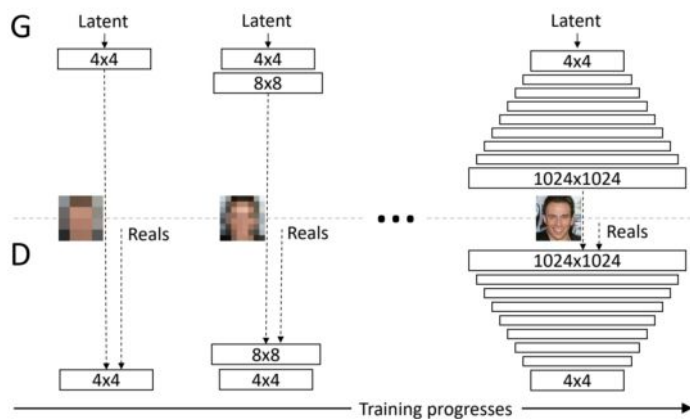


SEVERAL GANS

- More GAN-like models can be constructed by considering different divergences between probabilities
 - GANs build on Bergman divergences
 - Wasserstein GANs use the Wasserstein metric
- GANs and VAEs can be combined (VAE-GAN)
- Conditional GANs exist that work akin to continual VAEs

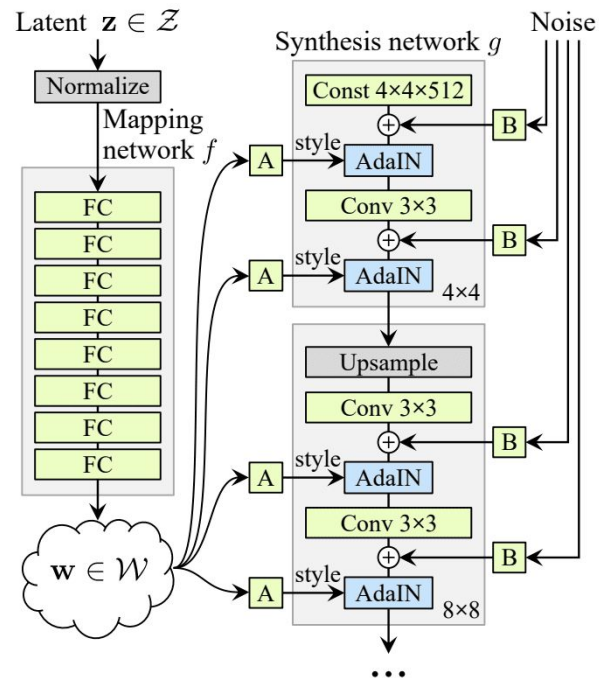
GENERATIVE MODELS TODAY

Progressive GAN



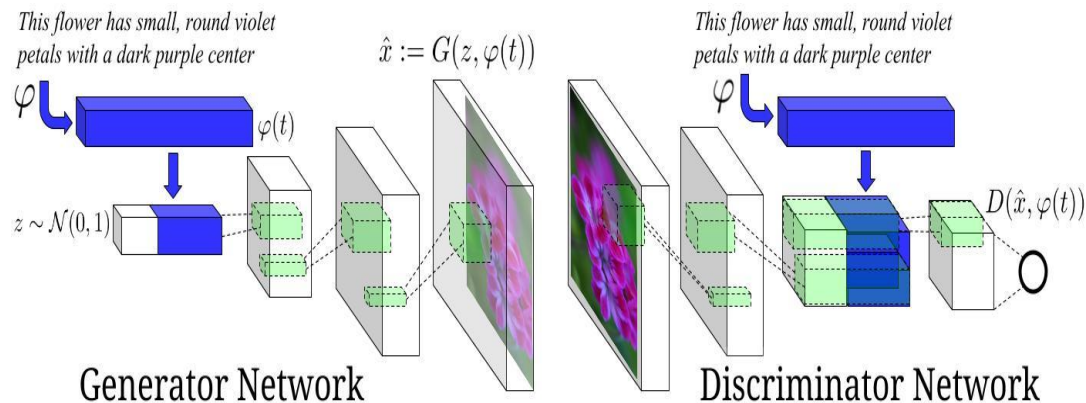
GENERATIVE MODELS TODAY

StyleGAN: incredibly realistic images



GENERATIVE MODELS TODAY

Multimodal data



this small bird has a pink
breast and crown, and black
primaries and secondaries.



the flower has petals that are bright pinkish purple with white stigma



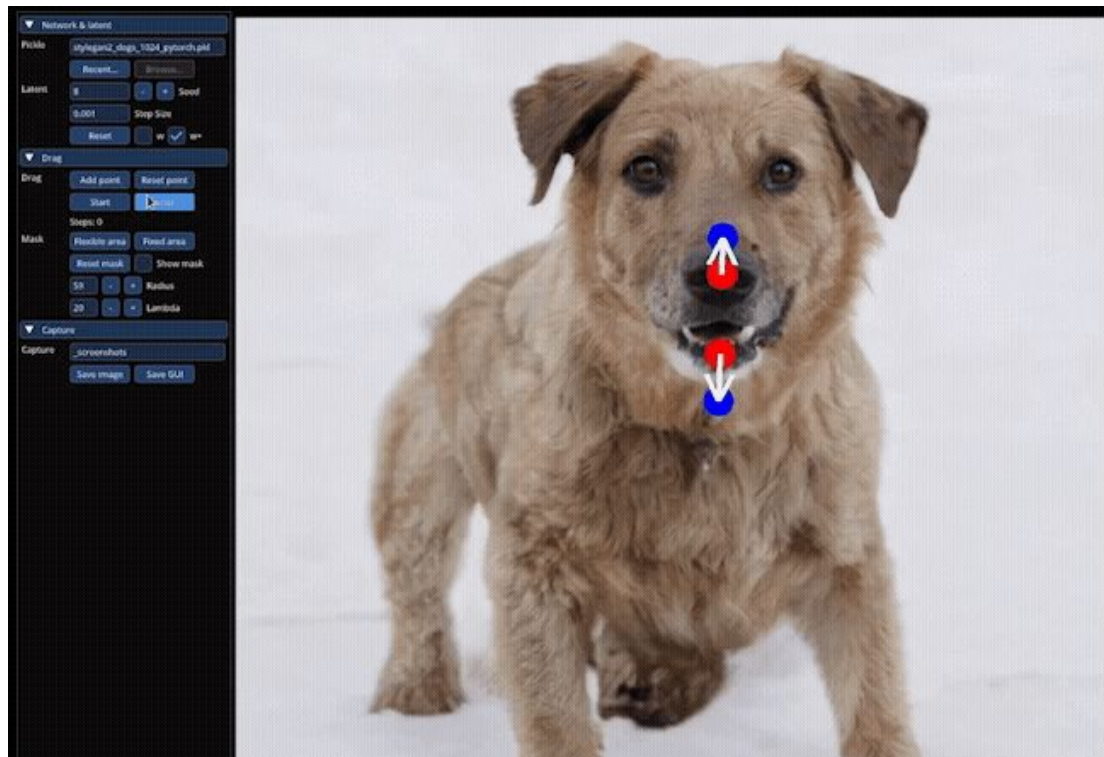
this magnificent fellow is almost all black with a red crest, and white cheek patch.



this white and yellow flower
have thin white petals and a
round yellow stamen



DRAG GAN



GENERATIVE MODELS

Video Generation



QUESTIONS?

